

James Sypherd

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PROFILE

Biomedical engineer, researcher, and game systems designer whose work spans clinical and orthopaedic research, motion capture and biomechanics, research instrumentation, real-time interactive systems, and applied AI. Experienced building gameplay in Unreal Engine 5 and Unity, designing behavioral and measurement hardware, and developing LLM-driven tools with prompt design, guardrails, and evaluation. Comfortable moving from study design and data analysis to shipping working software.

EDUCATION

Arizona State University - M.S., Biomedical Engineering, May 2025 (GPA 3.50)

Arizona State University - B.S.E., Biomedical Engineering, May 2024 (GPA 3.60)

RESEARCH EXPERIENCE

Phoenix Children's Hospital

Clinical Research Coordinator Fall 2025

- Coordinate clinical research studies, including participant enrollment, study documentation, and data collection and management.

Phoenix Children's Hospital

Orthopaedic Researcher 2023 - 2025

- Conducted orthopaedic and movement research supporting clinical motion analysis, data processing, and biomechanical assessment.
- Contributed to motion capture and gait analysis workflows in the Motion Analysis Laboratory.

Geometric Media Laboratory, Arizona State University

Graduate Researcher, Motion Capture Fall 2024 - Spring 2025

- Built MATLAB tools to import motion capture data and run inverse kinematics analysis on OptiTrack recordings.
- Designed capture and analysis workflows for studying timing, movement quality, and biomechanical variation.

Barrow Neurological Institute

Research Intern Summer 2021

- Designed and built a capacitance-based lickometer instrument for behavioral neuroscience experiments.
- Delivered a working device for licking-response measurement, programmed on Arduino (open-sourced as Cap-Lickometer).

TEACHING EXPERIENCE

Arizona State University

Professor of Practice, GAME School, and Graduate Teaching Assistant, AME 294 Spring 2025 - Present

- Teach a 3-credit course on motion capture, movement analysis, and real-time 3D visualization.
- Help students turn physical motion into readable interactive systems using OptiTrack workflows.
- Write instructional materials and technical documentation for creative technology pipelines.

ENGINEERING & GAME DEVELOPMENT EXPERIENCE

Endless Games and Learning Lab, Arizona State University

Software Engineer / Game Developer December 2025 - Present

- Develop gameplay systems and prototype features for educational and entertainment projects across Unity, Unreal Engine 5, UEFN, and Roblox.
- Implement player-facing mechanics, interaction systems, and rapid prototypes for combat, progression, and experiential design goals.
- Apply AI-assisted prototyping and tooling to move from design intent to working mechanics faster.
- Support motion capture and .c3d data workflows for gameplay, animation, and interactive media.
- Contribute Unreal Engine 5 work on multiple in-house projects, including gameplay prototyping and mechanic implementation.

Hunting Grounds Board Game Development Team

Lead Game Designer Spring 2023 - Summer 2023

- Led design of a modular tabletop game built around procedural learning through play.

- Built a prototype with 200+ cards, class-specific interactions, progression structures, and role-based mechanics.
- Designed itemization, rewards, and player guidance to strengthen replayability and class differentiation.

AI & LLM SYSTEMS

Applied AI Work (Independent / OpenClaw agent platform)

Builder / Systems Designer 2022 - Present

- Design system prompts, role-scoped behavior definitions, and safety guardrails that keep an autonomous agent in character and within policy.
- Built a multi-provider LLM routing layer across OpenAI, Anthropic, Google Gemini, and Groq, choosing models per task by capability, latency, and cost.
- Implemented retrieval over a structured memory and knowledge store so responses stay grounded in prior context, a working RAG pattern.
- Iterate on prompt versions and check for regressions when prompts or models change, an informal evaluation and quality practice.
- Wrote an institutional audit of AI tools for Arizona State University covering capability, data handling, and approved-use paths.
- Automate data workflows, scheduling, and tool integrations in Python.

SELECTED TECHNICAL PROJECTS

nAIVE Game Engine (Rust)

Contributor 2026

- Contribute to an open-source Rust game engine built on WebGPU and Vulkan with Lua scripting and SLANG shaders.
- Work includes an MCP server that exposes the engine to AI-driven control and tooling.

Rebuild Orchestrator (Unreal Engine 5.8 Plugin)

Author 2026

- Built an editor-only C++ plugin that runs a safe external clean, build, and relaunch loop through Unreal Build Tool, verified on the StrengthERG project.

Multi-Engine Game Prototypes

Solo Developer 2024 - Present

- Shipped prototypes across Unreal Engine 5, Unity, Roblox (Luau), and Godot, including Elementals (UE5), Baolf (Unity), Scientist Run and Brainrot Online (Roblox), and Thunderwalker 2D (Godot).

GAME DESIGN EXPERIENCE

The Blacksmith

Gameplay & Systems Developer 2025 - Present

- Build crafting and forging gameplay systems in Unreal Engine 5, including Blueprint-driven crafting and item-creation flows.
- Create and author enemy AI behavior trees, animation montages, blackboards, and AI controllers.
- Collaborate through a Perforce-managed multi-developer pipeline, coordinating streams, workspaces, and asset submissions.

Concept 2 Exercise Game Suite (StrengthERG)

Gameplay Engineer 2024 - Present

- Build exercise games in Unreal Engine 5.8 that integrate Concept 2 PM5 rowing and erg hardware in real time over TCP and CSAFE; led the suite's migration from an original Unity build.
- Design movement-driven mechanics that map stroke data and heart rate to in-game progression across rowing, cycling, and strength modes.

World of Warcraft / Final Fantasy XIV

MMO Systems Analysis 2016 - Present

- Study class design, talent structures, progression systems, and long-term reward loops across major expansion eras.
- Analyze how class fantasy, pacing, and combat feel shape player satisfaction and expression.

Dungeons & Dragons / Pathfinder

Dungeon Master and Systems Designer 2017 - 2025

- Ran weekly homebrew campaigns for 6 to 8 players with custom encounters, reward systems, and class

interactions.

- Created custom rules, classes, items, and conversion systems between D&D 5e and Pathfinder 2e.

Daggerheart

Playtester Spring 2024 - Summer 2024

- Selected for Daggerheart playtesting and delivered structured analysis grounded in play experience and mechanic breakdown.

SKILLS

Research: Study coordination, orthopaedic and biomechanics research, motion capture and gait analysis, research instrumentation and hardware, inverse kinematics, data analysis

AI / LLM: Prompt engineering, system prompt design, guardrails, RAG, LLM output evaluation, multi-provider LLM APIs (OpenAI, Anthropic, Google Gemini), agentic workflows

Programming: Python, C#, C++, Rust, Lua, MATLAB, Arduino / embedded, Git and GitHub, Perforce

Engines / Tools: Unreal Engine 5 (Blueprint + C++), Unreal gameplay AI (behavior trees, blackboards), UEFN, Unity, Roblox (Luau), Godot, OptiTrack, motion capture pipelines, .c3d workflows

Design: Systems design, combat and progression design, reward loops, class identity, encounter design, playtesting

CERTIFICATIONS & MICROCREDENTIALS

Endless Games and Learning Lab microcredential - Motion Capture for Martial Arts